

EXP.NO: 7	Develop an Android Calculator Application using Android Studio
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AIM:

To develop an Android application that implements a basic calculator using Android controls such as Button, TextView, and EditText. The calculator will support basic mathematical operations like Addition, Subtraction, Multiplication, and Division.

ALGORITHM:**1. Initialize the Android Application:**

- Set up a new Android project in **Android Studio**.
- Create a **MainActivity** Java or Kotlin file for the main screen of the application.

2. Design the User Interface (UI):

- Create the following UI components in the **XML layout** file:
 - **EditText** for input fields where the user will enter numbers (two EditTexts for number1 and number2).
 - **Buttons** for each arithmetic operation (Addition, Subtraction, Multiplication, Division).
 - **TextView** to display the result of the operation.

3. Declare Variables:

- Create variables for the **EditText** fields, **Buttons**, and **TextView**.
- Define variables to hold the input numbers and the result (e.g., num1, num2, result).

4. Get Input from the User:

- Get the values entered by the user in **EditText** fields using `getText().toString()` and convert them to numerical values (like `Integer.parseInt()` or `Double.parseDouble()`).

5. Set Event Listeners for Buttons:

- For each button (Addition, Subtraction, Multiplication, and Division), set an **OnClickListener**.
- Inside the **onClick()** method for each button, perform the respective arithmetic operation:

- **Addition:** $\text{result} = \text{num1} + \text{num2}$
- **Subtraction:** $\text{result} = \text{num1} - \text{num2}$
- **Multiplication:** $\text{result} = \text{num1} * \text{num2}$
- **Division:** $\text{result} = \text{num1} / \text{num2}$ (add validation to prevent division by zero).

6. Perform Arithmetic Operation:

- Check which button was clicked (using `view.getId()`), and based on the button, perform the respective arithmetic operation.
- **Handle Division by Zero:** Check if `num2` is 0 before performing division and show an appropriate error message if true.

7. Display the Result:

- Use **TextView** to display the result of the operation.
- Convert the result to a string and set it using `setText()` method on the **TextView**.

8. Validate User Inputs:

- Before performing any operation, validate the input to ensure the user has entered valid numbers (not empty or non-numeric characters).

9. Handle Edge Cases:

- Ensure the app handles cases where users input invalid values or attempt division by zero. Display appropriate messages for errors.

10. Finish the Operation:

- Once the result is displayed, the user can either perform another calculation or exit the app.

Code

Step 1: Create the Project

- **Template:** Empty Views Activity
- **Project name** – CalculatorApp
- **Package** - com.example.calculatorapp
- **Language:** Java
- **Minimum SDK** – API 24

- **Build system:** Groovy DSL

Step 2: Under res → layout → Updated activity_main.xml using **ConstraintLayout**

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <!-- First Number Input -->
    <EditText
        android:id="@+id/editTextNumber1"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:inputType="numberDecimal"
        android:hint="Enter first number"
        android:textSize="18sp"
        android:textColorHint="#000000"
        android:typeface="monospace"
        android:textStyle="bold"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        android:layout_margin="16dp" />

    <!-- Second Number Input -->
    <EditText
        android:id="@+id/editTextNumber2"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:inputType="numberDecimal"
        android:hint="Enter second number"
        android:textSize="18sp"
        android:textColorHint="#000000"
        android:typeface="monospace"
        android:textStyle="bold"
        app:layout_constraintTop_toBottomOf="@id/editTextNumber1"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        android:layout_margin="16dp" />
```

```
<!-- Row of Operation Buttons -->
<LinearLayout
    android:id="@+id/buttonRow"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:weightSum="4"
    android:layout_margin="16dp"
    app:layout_constraintTop_toBottomOf="@id/editTextNumber2"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent">

    <Button
        android:id="@+id/buttonAdd"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:text="Add" />

    <Button
        android:id="@+id/buttonSubtract"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:text="Subtract" />

    <Button
        android:id="@+id/buttonMultiply"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:text="Multiply" />

    <Button
        android:id="@+id/buttonDivide"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:text="Divide" />
</LinearLayout>

<!-- Clear Button -->
<Button
    android:id="@+id/buttonClear"
```

```
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Clear"
    app:layout_constraintTop_toBottomOf="@id/buttonRow"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    android:layout_marginTop="12dp"/>
```

```
<!-- Result TextView -->
```

```
<TextView
    android:id="@+id/textViewResult"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Result will appear here"
    android:textSize="18sp"
    android:textStyle="bold"
    app:layout_constraintTop_toBottomOf="@id/buttonClear"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    android:layout_marginTop="24dp" />
```

```
</androidx.constraintlayout.widget.ConstraintLayout>
```

Step 3 : Update the MainActivity.java file with this content

```
package com.example.calculatorapp;
```

```
import androidx.appcompat.app.AppCompatActivity;
```

```
import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
```

```
public class MainActivity extends AppCompatActivity {
```

```
    EditText editTextNumber1, editTextNumber2;
    Button buttonAdd, buttonSubtract, buttonMultiply, buttonDivide, buttonClear;
    TextView textViewResult;
```

```
    @Override
    protected void onCreate(Bundle savedInstanceState) {
```

```

super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);

// Link XML elements with Java code
editTextNumber1 = findViewById(R.id.editTextNumber1);
editTextNumber2 = findViewById(R.id.editTextNumber2);
buttonAdd = findViewById(R.id.buttonAdd);
buttonSubtract = findViewById(R.id.buttonSubtract);
buttonMultiply = findViewById(R.id.buttonMultiply);
buttonDivide = findViewById(R.id.buttonDivide);
buttonClear = findViewById(R.id.buttonClear);
textViewResult = findViewById(R.id.textViewResult);

// Add button logic
buttonAdd.setOnClickListener(view -> calculate('+'));

// Subtract button logic
buttonSubtract.setOnClickListener(view -> calculate('-'));

// Multiply button logic
buttonMultiply.setOnClickListener(view -> calculate('*'));

// Divide button logic
buttonDivide.setOnClickListener(view -> calculate('/'));

// Clear button logic
buttonClear.setOnClickListener(view -> {
    editTextNumber1.setText("");
    editTextNumber2.setText("");
    textViewResult.setText("Result will appear here");
});
}

private void calculate(char operator) {
    String input1 = editTextNumber1.getText().toString();
    String input2 = editTextNumber2.getText().toString();

    if (input1.isEmpty() || input2.isEmpty()) {
        textViewResult.setText("Please enter both numbers.");
        return;
    }

    try {
        double num1 = Double.parseDouble(input1);

```

```

double num2 = Double.parseDouble(input2);
double result = 0;

switch (operator) {
    case '+':
        result = num1 + num2;
        break;
    case '-':
        result = num1 - num2;
        break;
    case '*':
        result = num1 * num2;
        break;
    case '/':
        if (num2 == 0) {
            textViewResult.setText("Cannot divide by zero");
            return;
        }
        result = num1 / num2;
        break;
}

textViewResult.setText("Result: " + result);
} catch (NumberFormatException e) {
    textViewResult.setText("Invalid input");
}
}
}

```

Step 4 – Under Gradle Scripts section → update the build.gradle(Module:app)

```

plugins {
    alias(libs.plugins.androidApplication)
}

android {
    namespace 'com.example.calculatorapp'
    compileSdk 34

    defaultConfig {
        applicationId "com.example.calculatorapp"
        minSdk 26
        targetSdk 34
        versionCode 1
        versionName "1.0"
    }
}

```

```

        testInstrumentationRunner "androidx.test.runner.AndroidJUnitRunner"
    }

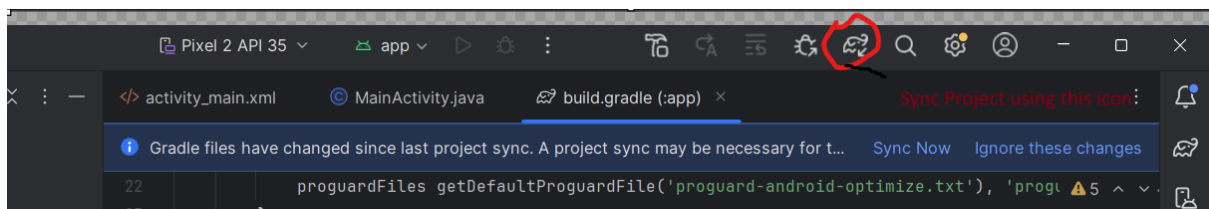
    buildTypes {
        release {
            minifyEnabled false
            proguardFiles getDefaultProguardFile('proguard-android-optimize.txt'), 'proguard-
rules.pro'
        }
    }
    compileOptions {
        sourceCompatibility JavaVersion.VERSION_1_8
        targetCompatibility JavaVersion.VERSION_1_8
    }
}

dependencies {
    // View/Layout
    implementation 'androidx.constraintlayout:constraintlayout:2.1.4'

    // Core libraries
    implementation 'androidx.core:core:1.12.0'
    implementation libs.appcompat
    implementation libs.material
    implementation libs.activity

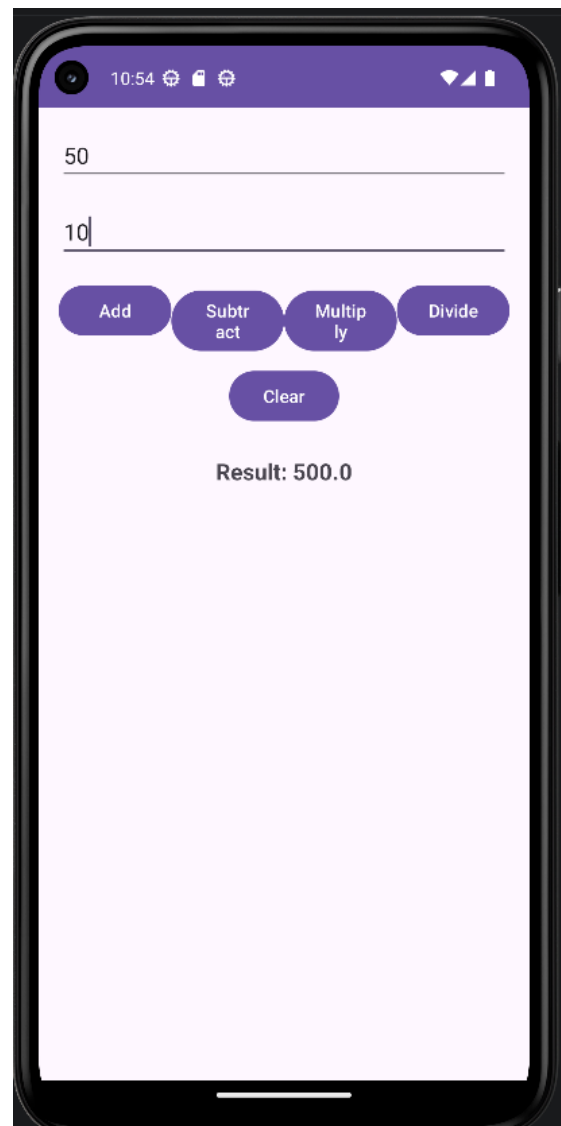
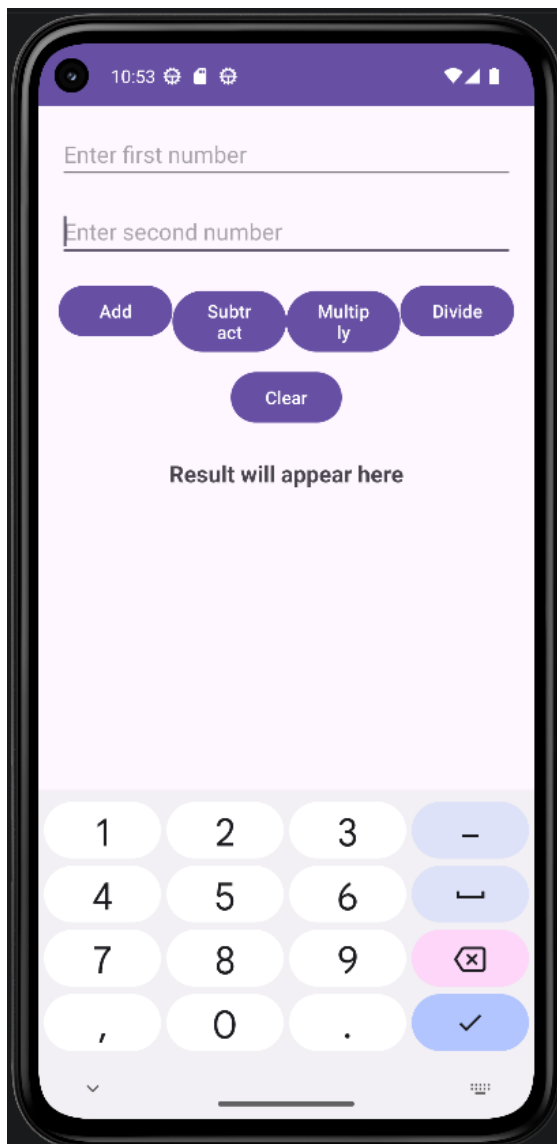
    // Testing
    testImplementation libs.junit
    androidTestImplementation libs.ext.junit
    androidTestImplementation libs.espresso.core
}

```



Step 5 – Click the Play Button to Run the project using the Emulator and verify the output as below

Output:



RESULT:

An calculator is designed using Android studio which uses controls like Button, TextView, EditText having basic functionality like Addition, Subtraction, multiplication, and Division.